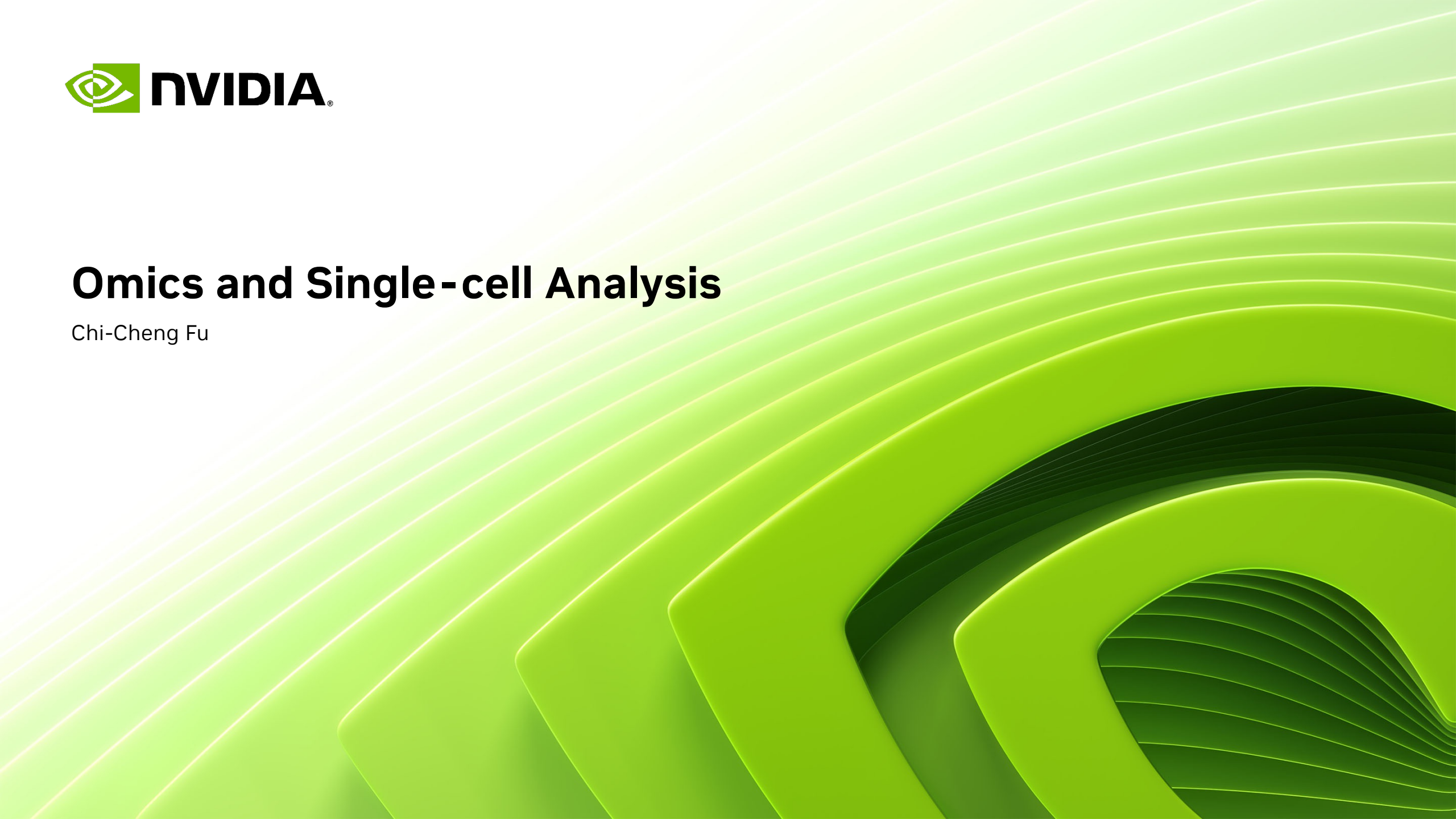


Omics and Single-cell Analysis

Chi-Cheng Fu



Agenda

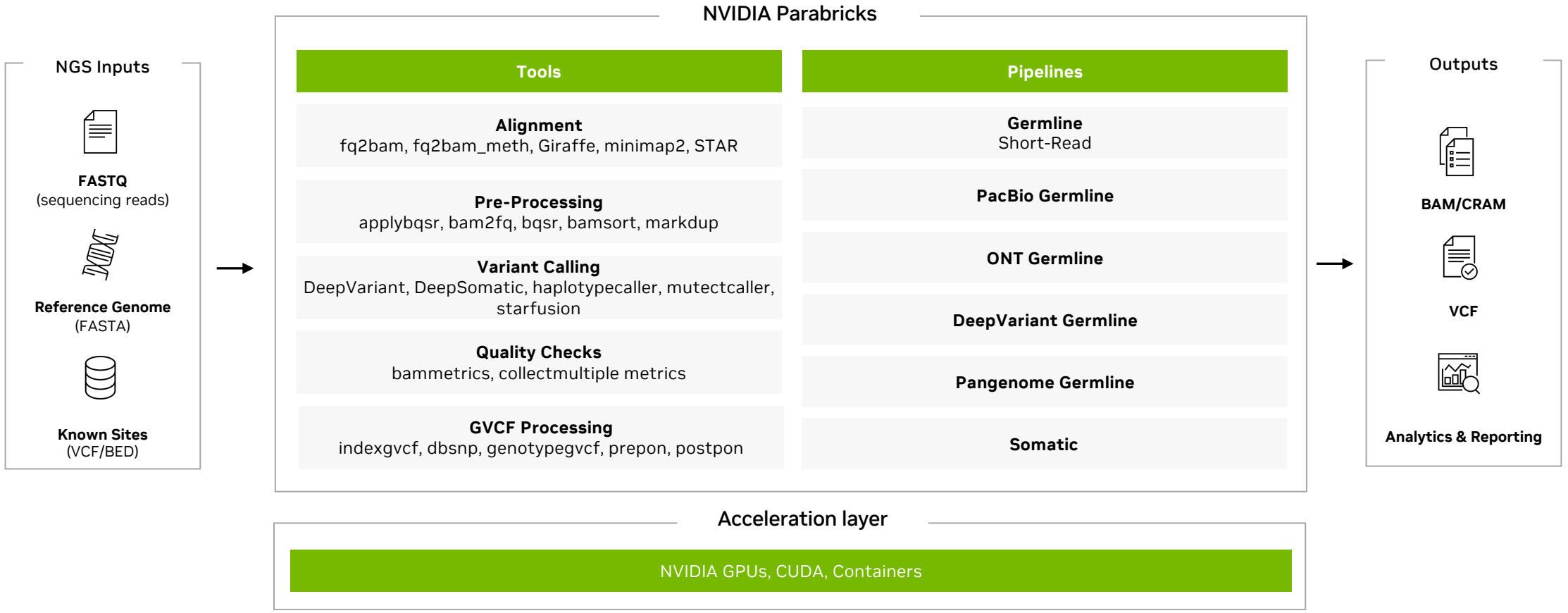
- Warm up notebook environment
- End-to-End Parabricks and CodonFM for Variant Effect Prediction
- Single-cell Analysis

NVIDIA Parabricks



NVIDIA Parabricks

GPU-Accelerated Software Suite for Secondary Analysis



NVIDIA Parabricks

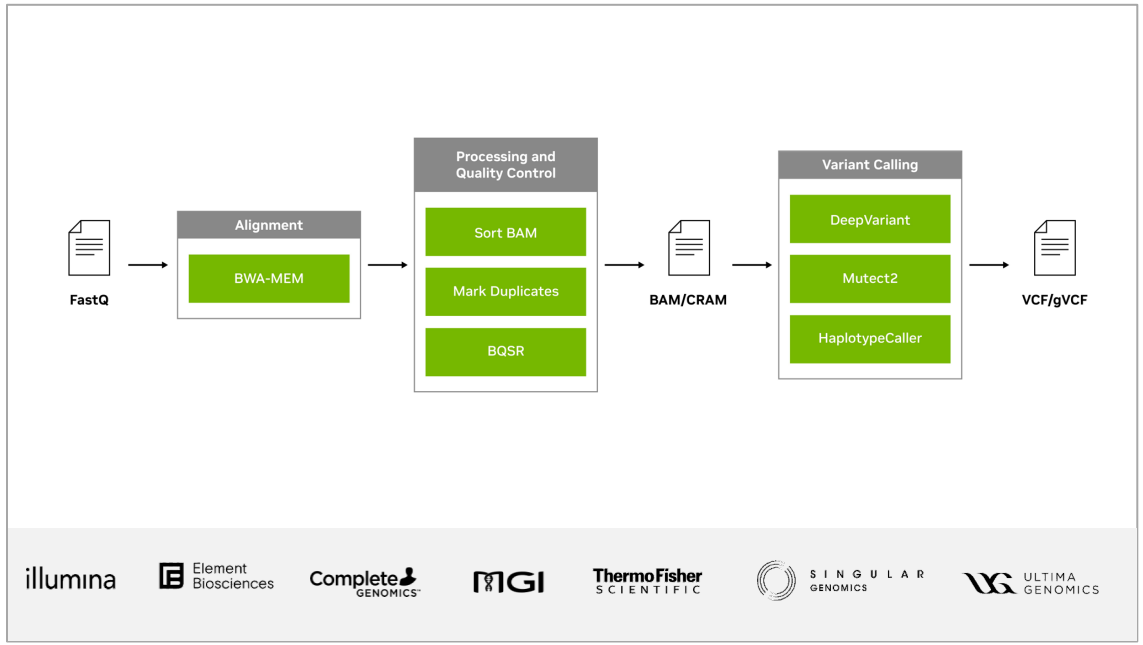
Tool Overview

Alignment	Pre-Processing	Variant Calling	Quality Checks	GVCF Processing
fq2bam (bwa-mem)	applybqsr	DeepVariant	bammetrics	indexgvcf
fq2bam_meth (bwa_meth)	bam2fq	DeepSomatic	collectmultiple metrics	dbsnp
Giraffe	bqsr	haplotypecaller		genotypevcf
Minimap2	bamsort	mutectcaller (Mutect2)		prepon
rna_fq2bam (STAR)	markdup	starfusion		postpon

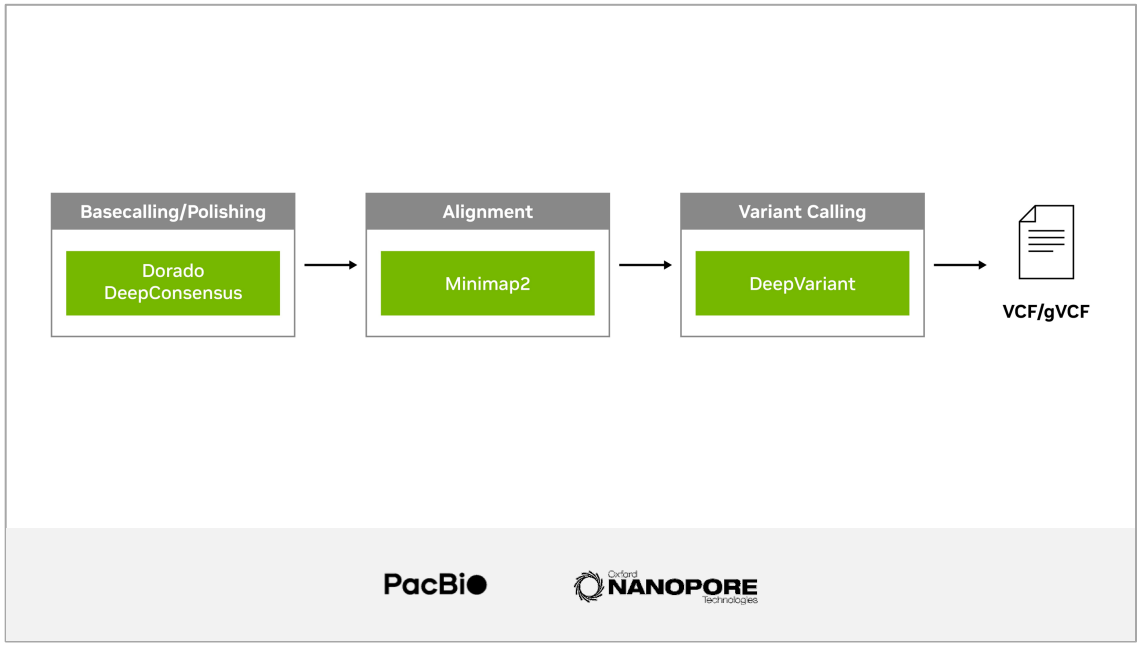
A Universal Analysis Solution

Accelerated analysis for both short-read and long-read sequencing

Short-Read



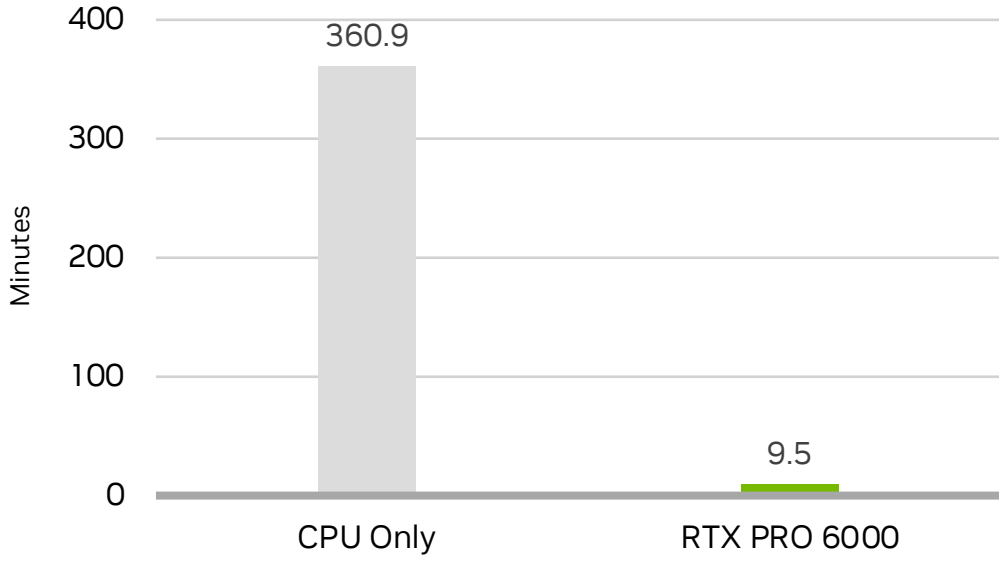
Long-Read

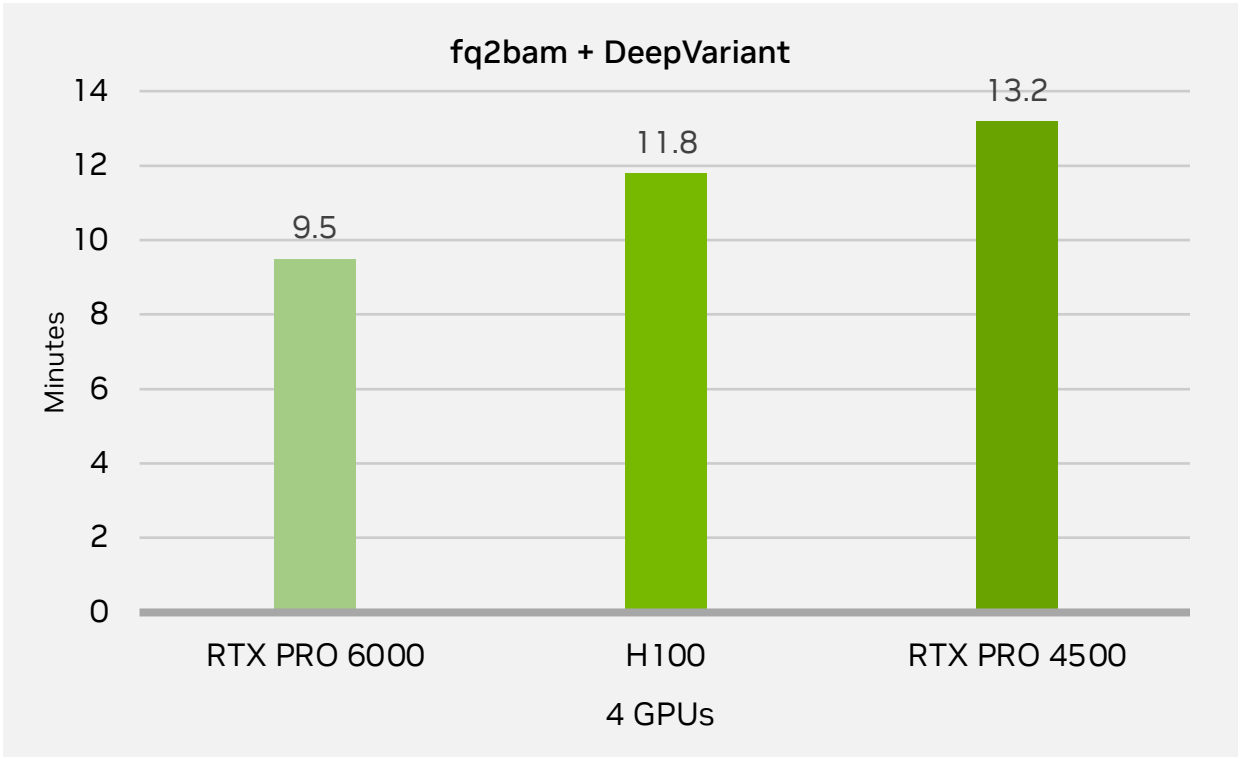


NVIDIA

Under 10-Minute Germline Analysis

Reducing germline analysis from hours to minutes





CPU instance: Intel Xeon with 96 vCPU (m5.24xlarge)

Accelerated Genomic Analysis

NVIDIA Parabricks provides equivalent results to open-source, gold standard tooling in a fraction of the time



Increase Speed

Under 10-minute germline analysis with the NVIDIA RTX PRO 6000



Reduce Cost

Lower compute costs up to 50% for WGS compared to CPU-only solutions



Enhance Accuracy

Deploy high-accuracy deep learning for all major sequencers



Ensure Transparency

Reproduce results from trusted, open-source tools

Variant Effect Prediction

How do variants in the DNA sequence affect the final protein?

	DNA	RNA	Amino Acid	Protein
Reference	**CTAAGCGGT**	**CUA AGC GGU**	**Leu-Ser-Gly**	
Sample #1	**CTAAGAGGT**	**CUA AGA GGU**	**Leu-Arg-Gly**	
Sample #2	**CTAAGTGGT**	**CUA AGU GGU**	**Leu-Ser-Gly**	

Variant

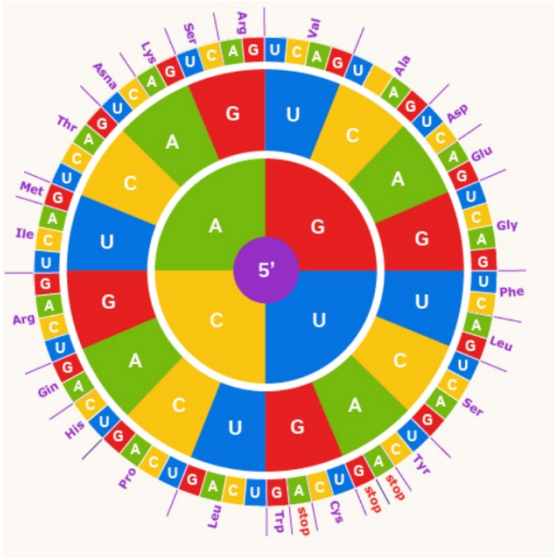
Synonymous mutation

NVIDIA

CodonFM is Largest Foundation Model to Address “Silent” Mutations

Unlike existing protein foundation models that treat synonymous codons interchangeably, codonFM provides codon-level resolution instead of treating amino acids as tokens

Image from Arc Institute's *The Hidden "Grammar" Revealed by CodonFM, Our Foundation Model for Codons*



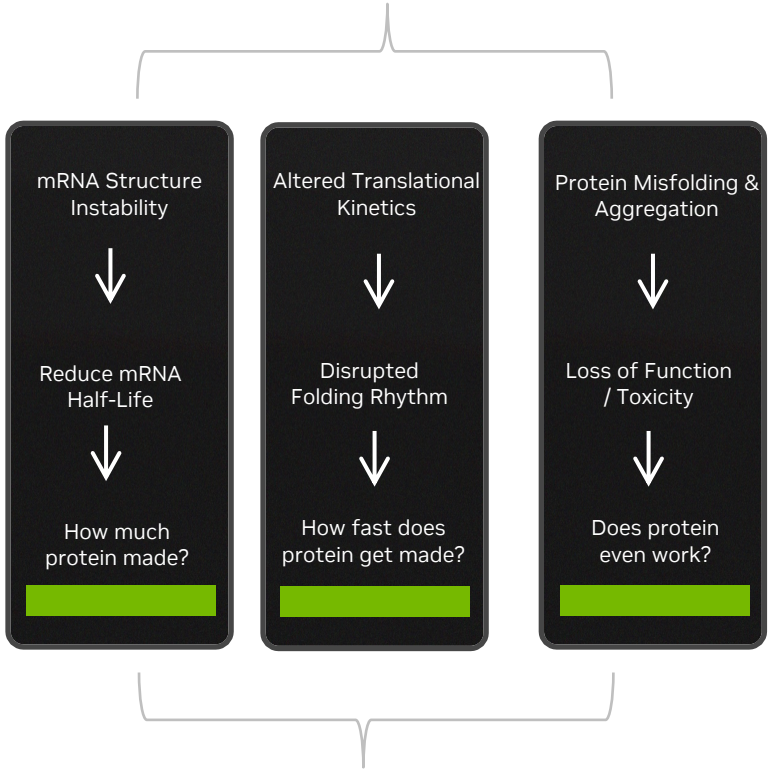
Provide Codon Level Insights

Existing Protein Foundation Models ignore codon-level detail and treatment synonymous codons interchangeably (ie, look at amino acid: Leu)

CodonFM is the largest foundation model to provide codon resolution instead treating amino acids as tokens (ie, look at CUA or CUG instead of Leu)

Synonymous Mutations

I.e, CUA and CUG are both Leu amino acid



Synonymous Lead to Silent Mutations

Upstream mutations can cause downstream consequences that impact disease detection and treatment

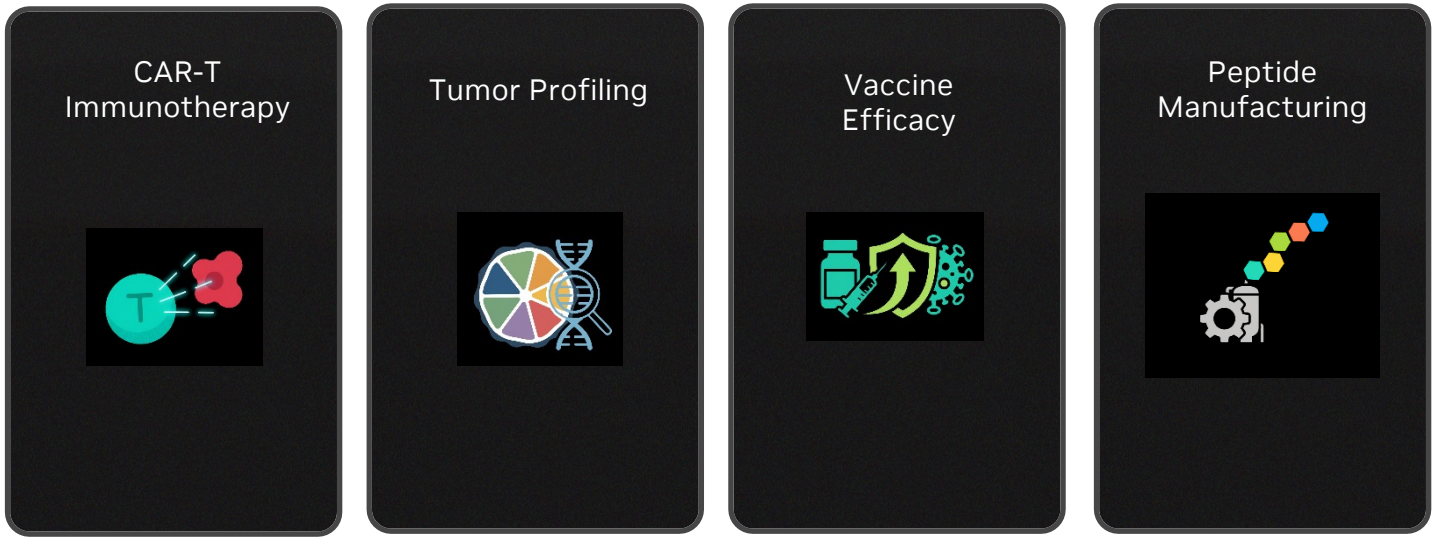
CodonFM 1 Billion Parameter Enhances Disease Detection and Treatment

Reduce downstream consequences of “silent” mutations—including mRNA structure instability, altered translational kinetics, and protein misfolding



CodonFM

Optimizes RNA sequences



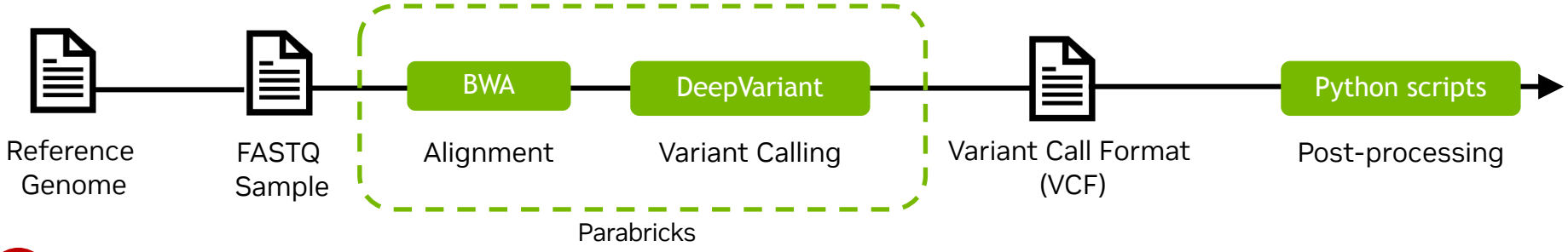
Understanding the Role of Codon Choices

The 5 billion parameter reduces downstream consequences of “silent mutations” to enhance accuracy and performance—resulting in improved disease detection and treatment

Variant Effect Prediction

Notebook overview

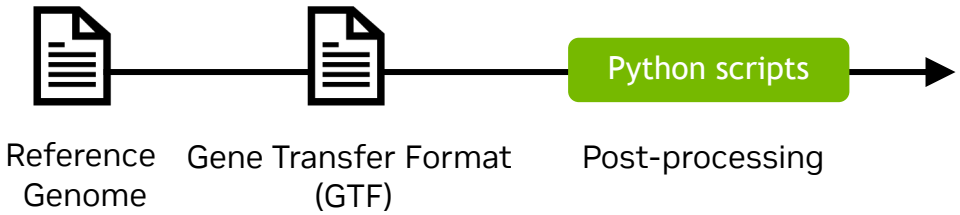
1 Variant Calling



Chrom	Pos	Ref	Alt
Chr1	16378	T	C
Chr1	63268	T	C
...	...	...	...

Variant Table

2 Gene Annotation



Gene Name	CDS Start	CDS End	CDS Sequence
OR4F5	65564	70008	ATGAAGAAGG...
OR4F16	73268	944153	ACAGCGTTCC...
...	...	...	...

Gene Annotations Table

3 Variant Effect Prediction

Gene Name	Ref Codon	Alt Codon	Ref Sequence
OR4F5	CGT	CCT	ATGAAGAAGG...
OR4F16	TCA	TCG	ACAGCGTTCC...
...	...	...	...

CodonFM Input



Variant-Effect Prediction

Gene Name	Log Likelihood
OR4F5	24.886238
OR4F16	23.277592
...	...

CodonFM Output

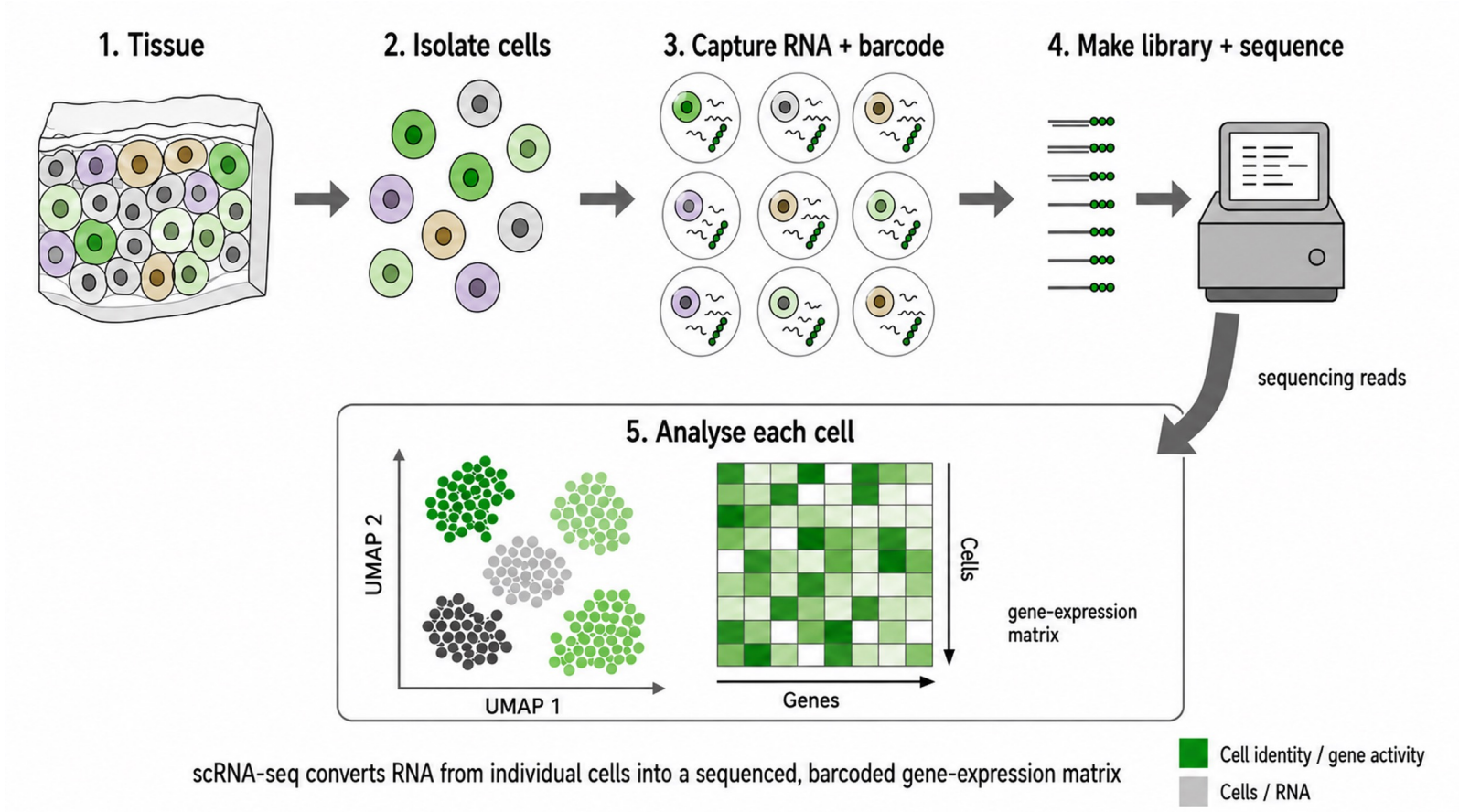
NVIDIA

Single-Cell Analysis



Single-Cell RNA Sequencing (scRNA-seq)

From mixed tissue to a cell-by-gene expression map



NVIDIA

CUDA-X

GPU DATA SCIENCE

ACCELERATED DATA SCIENCE

The RAPIDS suite of open source software libraries gives you the freedom to execute end-to-end data science and analytics pipelines entirely on GPUs.

Learn about RAPIDS »

SCALE OUT ON GPUS

Seamlessly scale from GPU workstations to multi-GPU servers and multi-node clusters with Dask.

Learn about Dask »

PYTHON INTEGRATION

Accelerate your Python data science toolchain with minimal code changes and no new tools to learn.

Learn about our libraries »

TOP MODEL ACCURACY

Increase machine learning model accuracy by iterating on models faster and deploying them more frequently.

Learn about RAPIDS for model optimization »

REDUCED TRAINING TIME

Drastically improve your productivity with more interactive data science tools like XGBoost.

Learn about XGBoost »

Learn about accelerated ML with cuML »

OPEN SOURCE

RAPIDS is an open source project. Supported by NVIDIA, it also relies on Numba, Apache Arrow, and many more open source projects.

Learn about our projects »

NVIDIA.

cuDF

A GPU DataFrame library in Python with a pandas-like API built into the PyData ecosystem

CPU

pandas

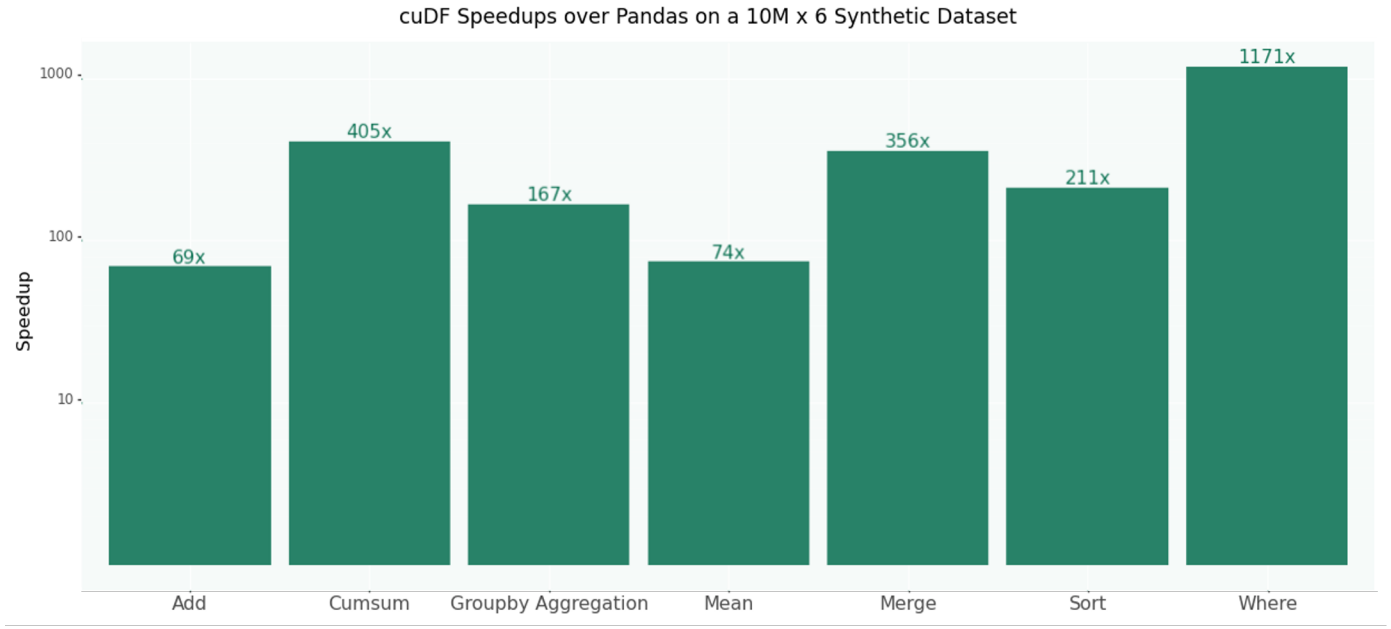
```
>>> import pandas as pd
>>> df = pd.read_csv("filepath")
>>> df.groupby("col").mean()
>>> df.rolling(window=3).sum()
```

GPU

cuDF

```
>>> import cudf
>>> df = cudf.read_csv("filepath")
>>> df.groupby("col").mean()
>>> df.rolling(window=3).sum()
```

Average Speed-Ups: *10-100x*



NVIDIA A100 vs. AMD EPYC 7642 48-Core Processor

Groupby	Strings and Regex	UDFs	Nested Types	Time Series
Indexing	Missing Data	CuPy Interoperability	Rolling Windows	

cuML

Accelerated machine learning with a scikit-learn API

CPU

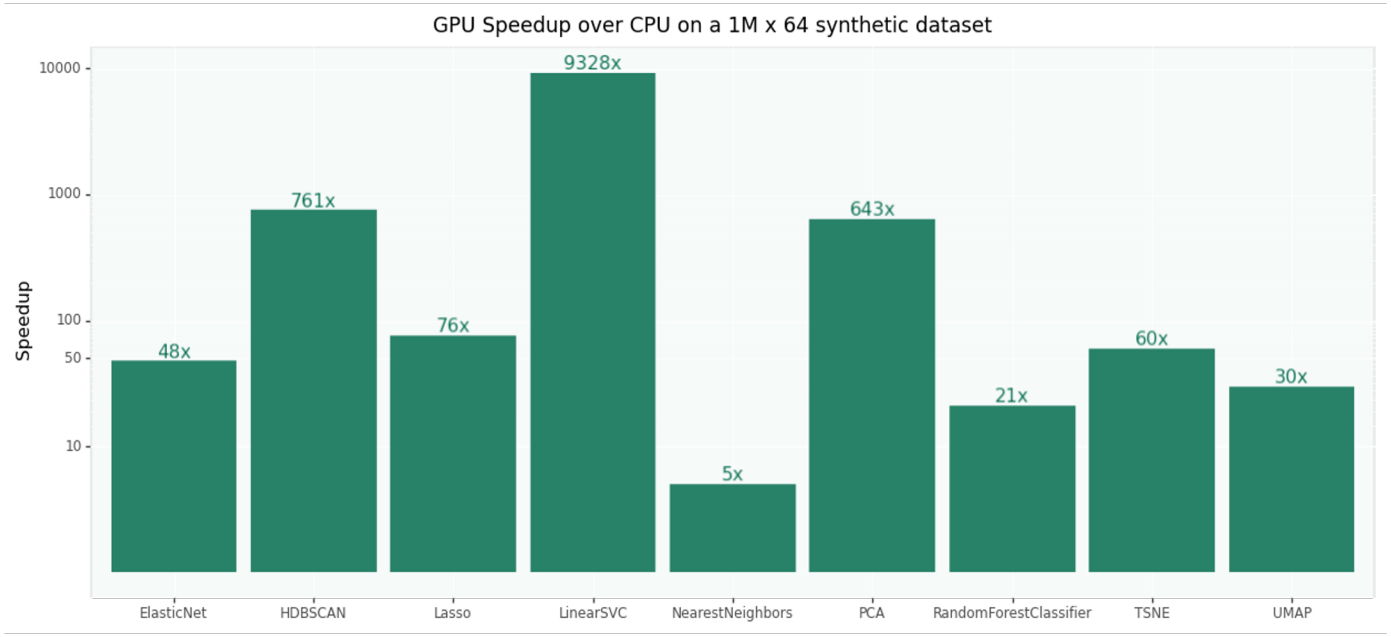
Scikit-learn

```
>>> from sklearn.ensemble import
RandomForestClassifier
>>> clf = RandomForestClassifier()
>>> clf.fit(x, y)
```

GPU

cuML

```
>>> from cuml.ensemble import
RandomForestClassifier
>>> clf = RandomForestClassifier()
>>> clf.fit(x, y)
```

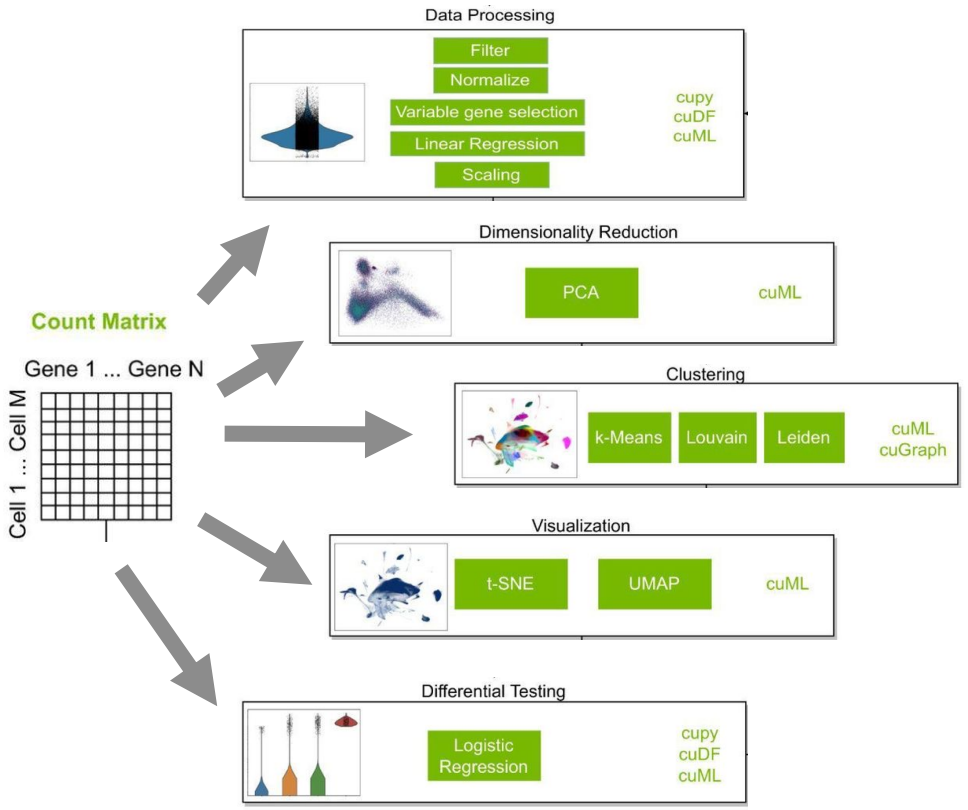


NVIDIA A100 vs. AMD EPYC 7642 96-Core Processor

Time Series	Classification	Regression	Clustering	Preprocessing
Cross Validation	Tree Models	Dimensionality Reduction	Explainability	

Single-Cell Analysis

Accelerated processing, clustering and visualization



https://www.biorxiv.org/content/10.1101/2022.05.26.493607v1.full

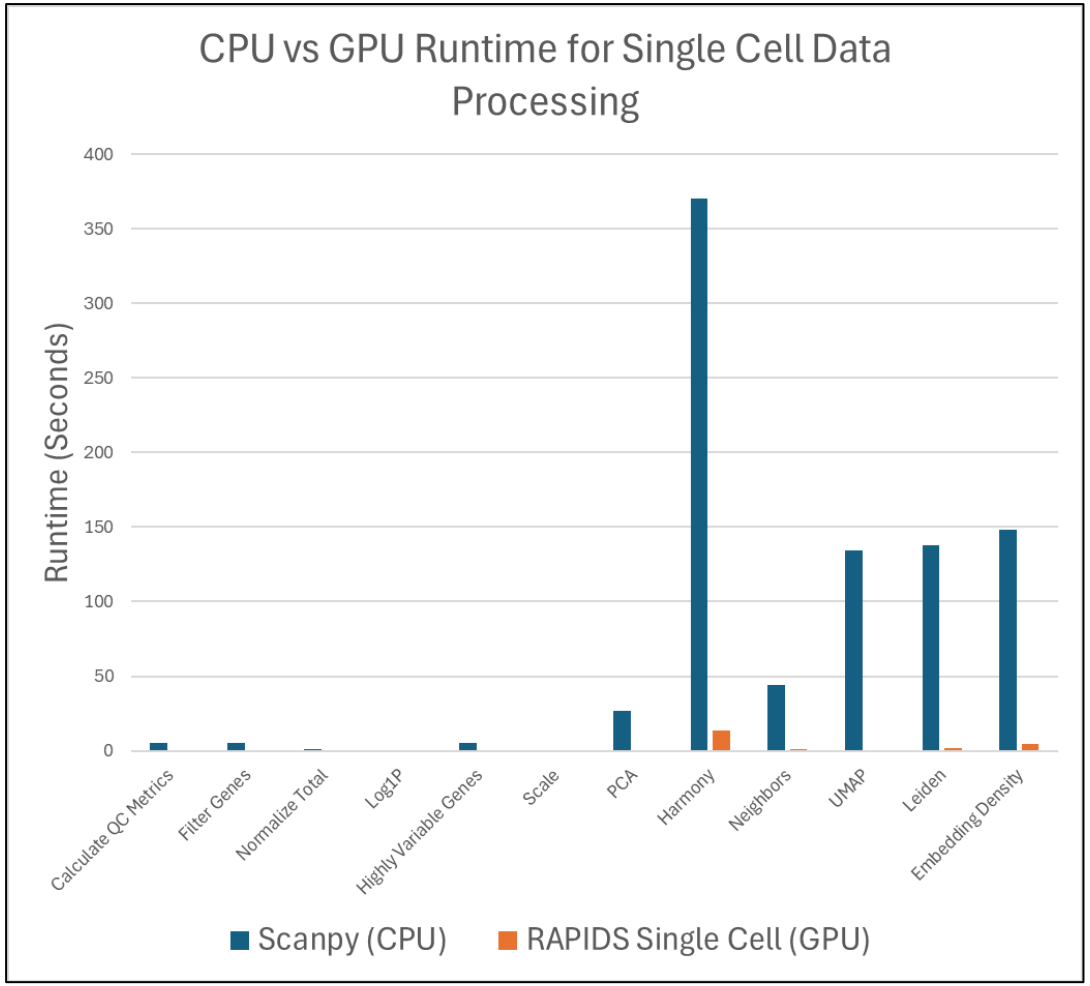
- Single-cell tertiary analysis relies heavily on regression, clustering and visualization methods
- All can be accelerated on GPU as part of the CuPy and RAPIDS ecosystem
- For 1.3 million cells, end-to-end runtime of all the processes on the left can be accelerated **from 4.3 hours on 40 x vCPU, to 14 mins on a single A100 GPU**

Accelerate Data Science and Leverage Foundation Models in Digital Biology

Scanpy (CPU) vs RAPIDS Single Cell (GPU) runtime performance

Calculation	Scanpy (CPU)	RAPIDS Single Cell (GPU)	Speedup
Calculate QC Metrics	5.61	0.098	57
Filter Genes	4.97	0.242	21
Normalize Total	0.945	0.0757	12
Log1P	0.342	0.0119	29
Highly Variable Genes	4.99	0.244	20
Scale	0.58	0.00513	113
PCA	27	0.614	44
Harmony	370	13.4	28
Neighbors	44.2	1.26	35
UMAP	134	0.39	344
Leiden	138	1.65	84
Embedding Density	148	4.54	33

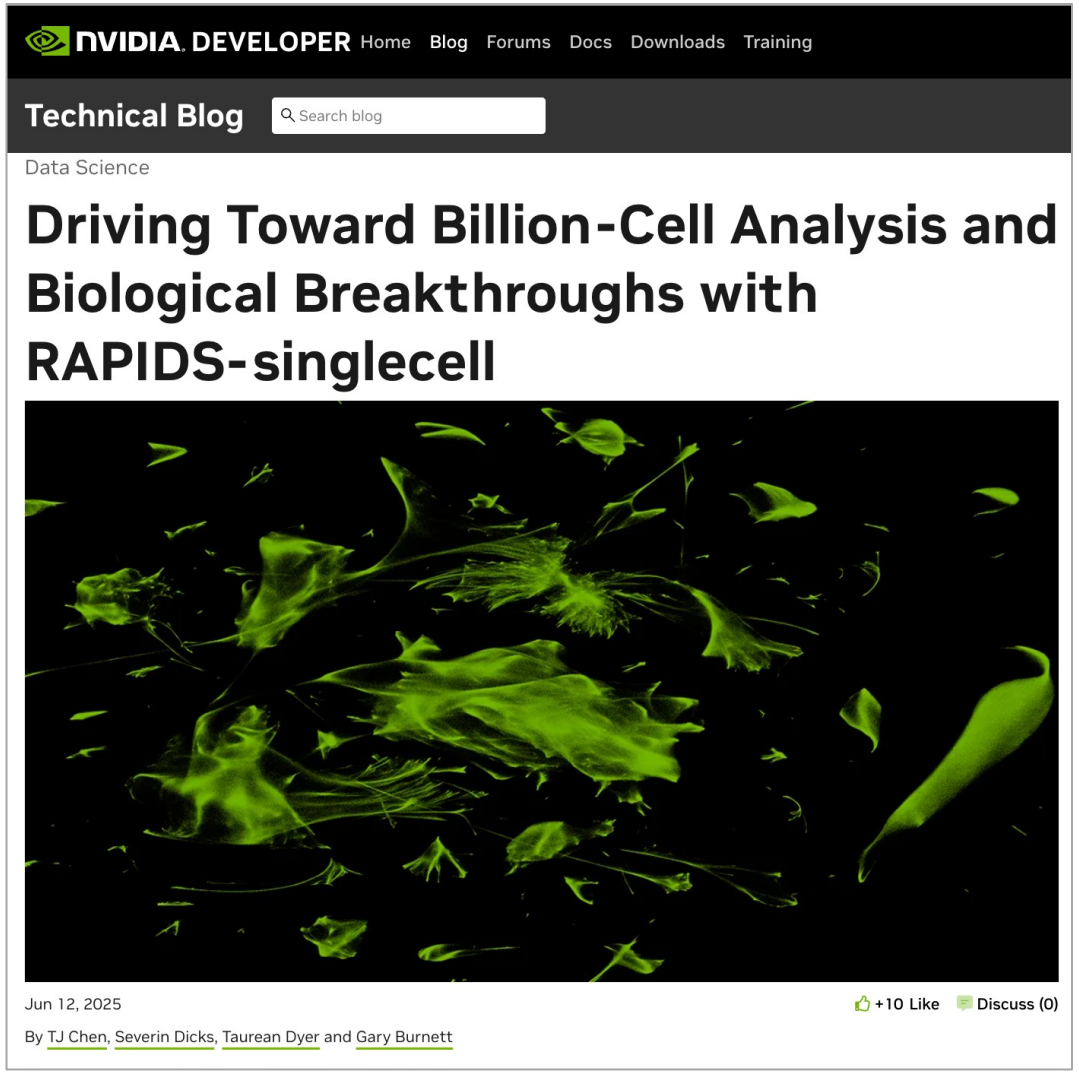
Runtime in seconds



Single-Cell Analysis

Included with NVIDIA AI Enterprise

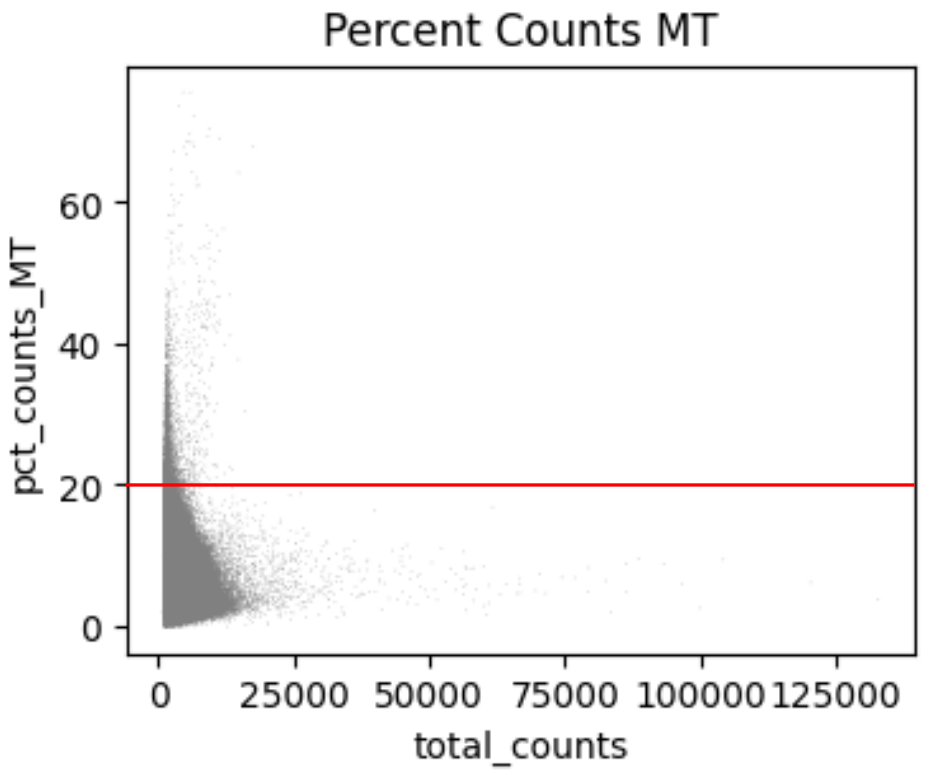
- A new library drawing inspiration from the rapids-single-cell-examples library and the scverse ecosystem
- Introduces GPU-optimized versions of scanPy (single cell) and squidPy (spatial) functions
- The library's primary objective is to blend the computation strength of GPUs with the user-friendly nature of the scverse ecosystem

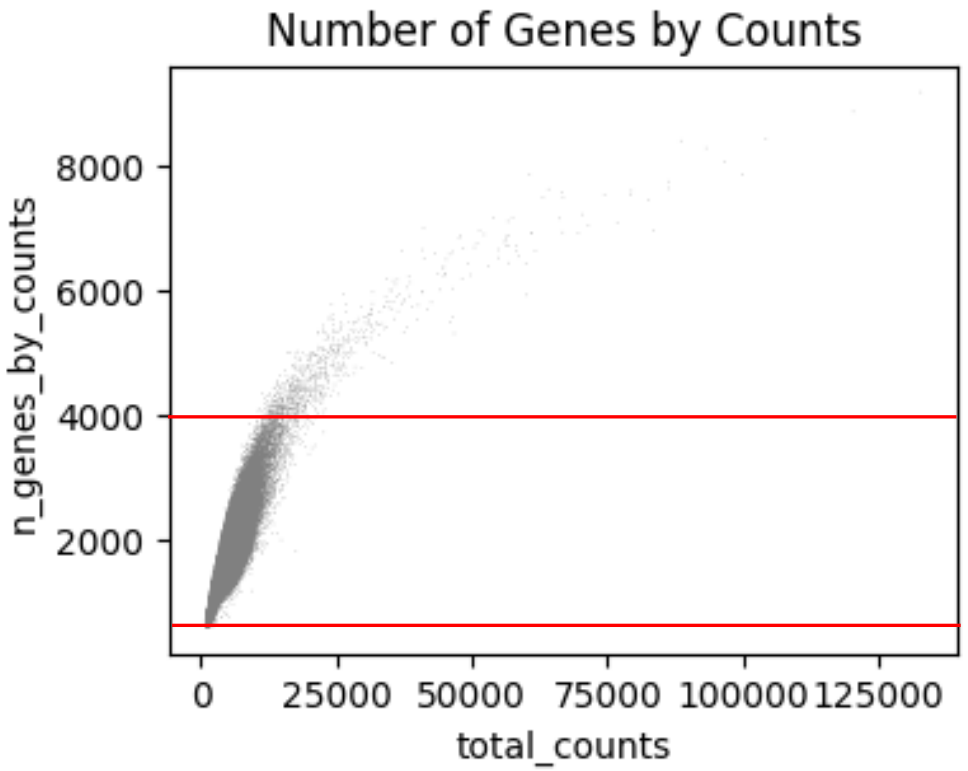


Link to technical blog

Accelerate Data Science and Leverage Foundation Models in Digital Biology

Single Cell Data Processing – QC Metrics and Filtering





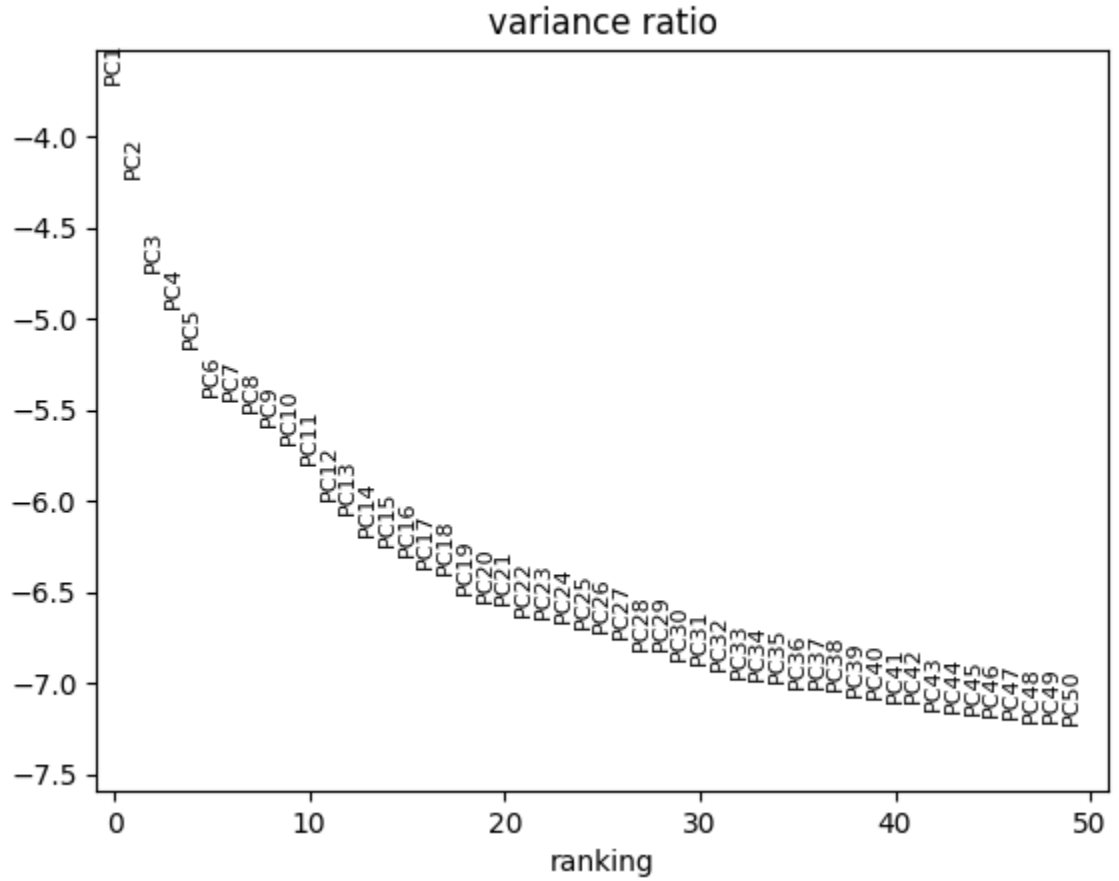
Removing cells with too many mitochondrial genes and too many or too few total genes

Accelerate Data Science and Leverage Foundation Models in Digital Biology

Single Cell Data Processing – Normalization, Feature Selection, PCA

Dimensionality reduction from 60k to 50

1. Normalize the variance of counts
2. Select highly variable genes
3. Run PCA for dimensionality reduction

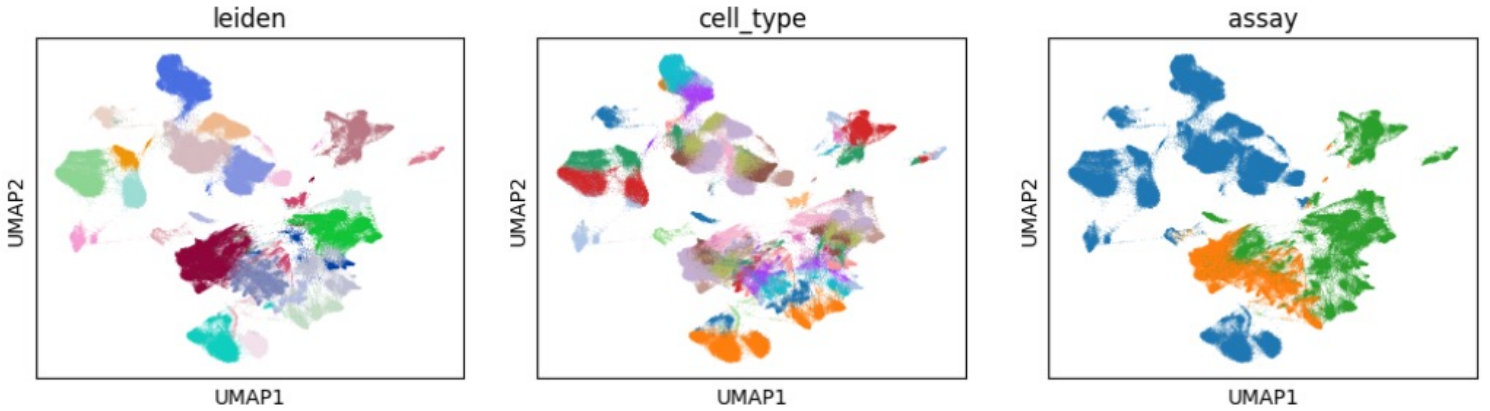


Plotting 50 principal components

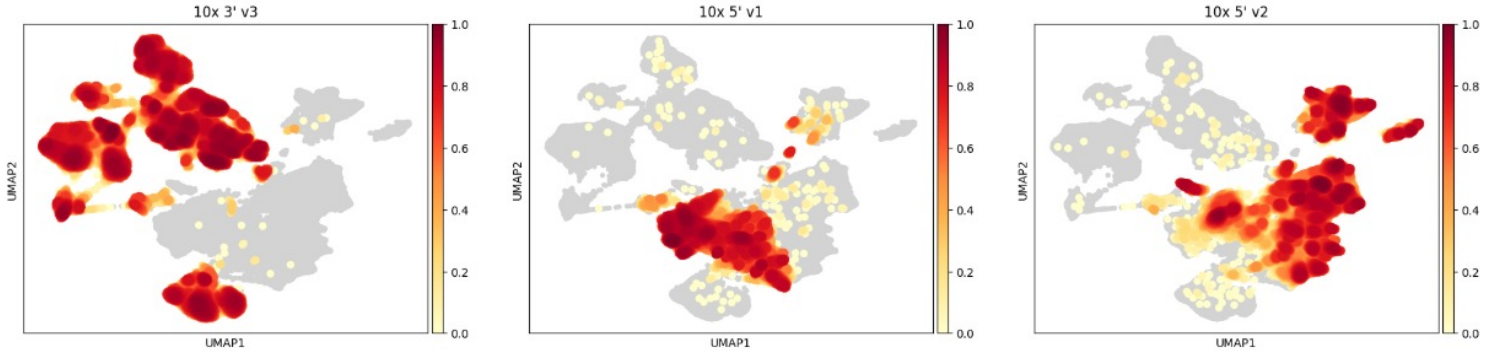
Accelerate Data Science and Leverage Foundation Models in Digital Biology

Single Cell Data Processing – Clustering and Embedding Density

Leiden clustering

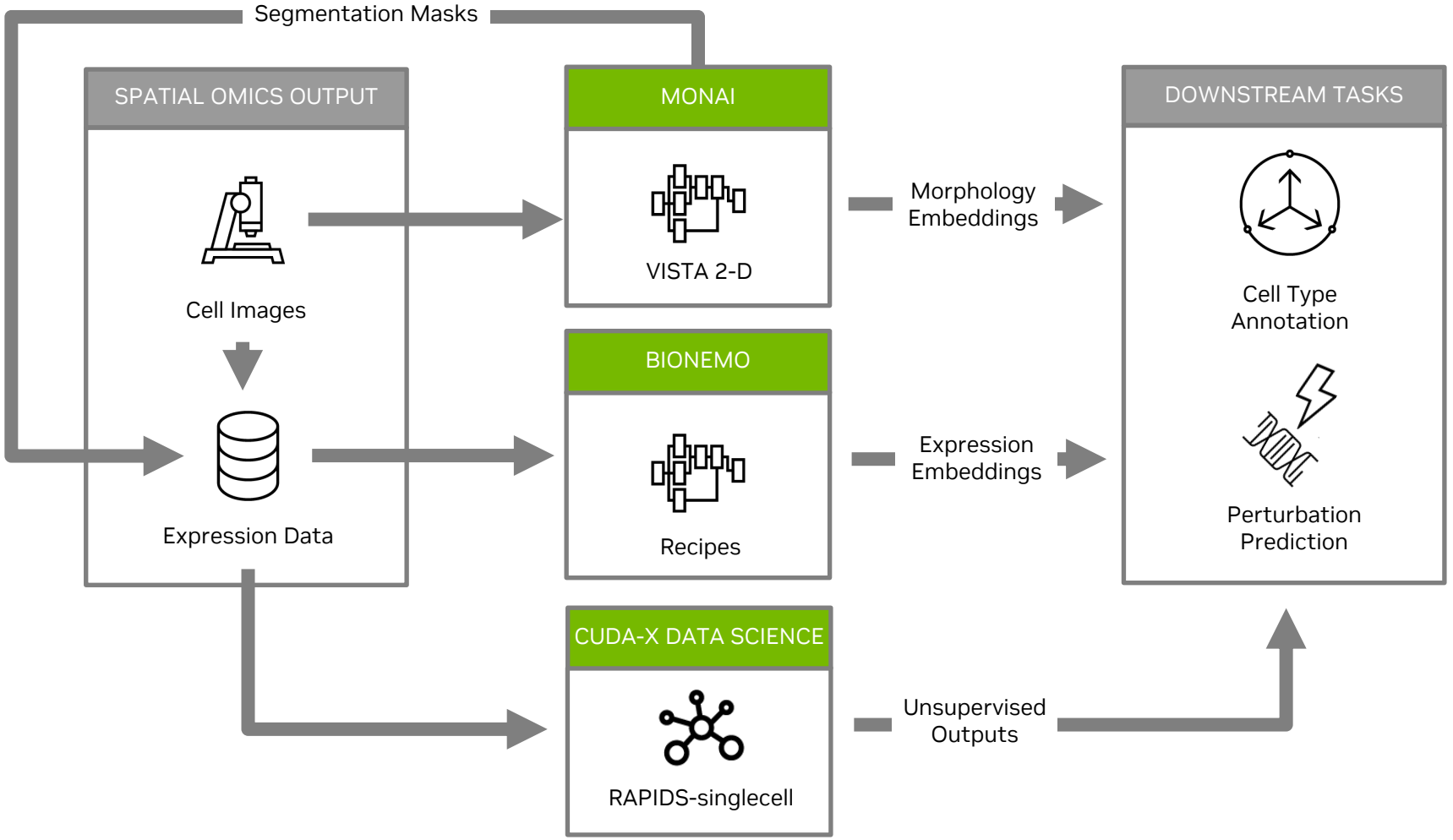


Embedding density



AI-Powered Single Cell & Spatial Omics

Build foundation models | Segment at high accuracy | Extract morphology embeddings



Helpful Resources

Links to dive deeper into the content

Parabricks Documentation – Discover the full set of Parabricks tools
CodonFM Github – Dive into the CodonFM code
CodonFM Blog – Learn more about the capabilities of the model

RAPIDS-singlecell Documentation – Learn about all the capabilities of RAPIDS-singlecell
RAPIDS-singlecell Github – Explore the code behind RAPISD-singlecell
RAPIDS-singlecell Paper – Understand how it works over the end
Scverse Zulip Chat –Join the Scverse community on Zulip